

Buğra Kaplan

Mechanical Engineer · Civil Engineer · Research Assistant | PhD Candidate in Mechanical Engineering

Işık University, Department of Mechanical Engineering · Istanbul Technical University, doctoral research

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ACADEMIC PROFILE

Alongside my degree in Mechanical Engineering, I completed a second bachelor's degree in Civil Engineering through a double-major programme. I am a research assistant in the Department of Mechanical Engineering at Işık University. I am currently at the dissertation stage of my PhD in Mechanical Engineering at Istanbul Technical University, where I study wind turbine performance using physics-based modelling and machine learning.

EDUCATION

PhD in Mechanical Engineering

Istanbul Technical University | 2023-present | Dissertation stage

I am at the dissertation stage of the PhD in Mechanical Engineering. My research examines wind-farm performance and early indications of turbine faults. I combine SCADA data, physics-based modelling and machine learning, and investigate turbine wakes and the contribution of acoustic data. GPA: 3.81/4.00.

MSc in Mechanical Engineering, Thermal and Fluid Sciences

Istanbul Technical University | 2020-2023 | First in programme, high honours

In my master's thesis, I studied machine-learning methods for forecasting wind-turbine power generation and developed an approach combining model outputs. I completed the programme with high honours, ranked first, with a GPA of 3.69/4.00.

BSc in Civil Engineering, double major

Işık University | 2016-2019 | Full scholarship, first in department

Alongside Mechanical Engineering, I completed the Civil Engineering double-major programme on a full scholarship. I graduated first in the department with a GPA of 3.87/4.00.

BSc in Mechanical Engineering

Işık University | 2014-2018 | Full scholarship, first in department and Faculty of Engineering

I studied Mechanical Engineering on a full scholarship. With a GPA of 3.85/4.00, I graduated first in both the department and the Faculty of Engineering.

ACADEMIC AND PROFESSIONAL EXPERIENCE

Doctoral Researcher, SÜİT Programme

TÜBİTAK | 2023-2026

In SÜİT, I worked on energy and communication technologies for sustainable urban infrastructure. Within energy-efficient IoT communication, I contributed to work that transmitted data packets over an optical channel using visible light. I also researched low-carbon generation, smart mobility and building energy efficiency. In interdisciplinary teams, I worked on design and analysis and contributed to scientific publications.

Research Assistant, Department of Mechanical Engineering

Işık University | 2019-present

At Işık University, I lead the theoretical and practical teaching of undergraduate and graduate courses for which I have academic responsibility and advise students. My research work includes experimental data collection, numerical modelling and analysis; I contribute to national and international projects and peer-reviewed publications. I also take part in coordinating departmental courses and examinations.

Software, Database and ERP Systems Intern Engineer

Bahçeşehir University and Wissen Akademi | 2019-2020

I developed applications with C# .NET, ASP.NET MVC, AngularJS and JavaScript. I worked on database design and data access with MS SQL, T-SQL, Entity Framework and LINQ. I also contributed to functional developments for business processes in Dynamics 365 AX, CRM and Finance & Operations.

Academic Assistant

Işık University | 2017-2019

As an academic assistant at Işık University, I supported course preparation, examinations and students' work. I contributed to faculty research projects and the preparation of scientific publications. I also participated in departmental academic meetings and the coordination of educational activities.

R&D Engineer Intern

Dalgakıran Compressor | June-July 2017

I took part in design, performance testing and technical analysis for new product development. I worked with R&D and production teams to transfer design results into manufacturing and contributed to assessments of cost, efficiency and service life for existing products.

Production Engineer Intern

CNK Aviation Industry | May-June 2016

I examined machine and process performance, quality-control results and efficiency indicators on the production line. I took part in transferring product designs into manufacturing and assessing production feasibility. Statistical reports on material quality, scrap rates and time use supported process-improvement work.

SELECTED RESEARCH AND ENGINEERING PROJECTS

Wind-farm performance and early fault diagnosis

Doctoral research · ongoing

In my doctoral research, I examine wind-farm performance and fault risk through a layered approach combining physical principles and machine learning. Alongside SCADA records, I account for the thermodynamic effects of air density, temperature, pressure and humidity in power models. I study whether deviations between physics-based predictions and measured output indicate performance loss or early faults. At the farm scale, I analyse layout- and wind-direction-dependent wake effects with graph neural networks. I also aim to test whether combining turbine acoustics with operational data improves performance forecasting and fault diagnosis.

SÜİT: Advanced Technologies for Sustainable Cities

TÜBİTAK · 2023-2026

As a doctoral researcher in the TÜBİTAK-supported SÜİT programme, I worked on energy and communication technologies for sustainable urban infrastructure. Within energy-efficient IoT communication, we successfully transmitted data packets over an optical channel using visible light communication (VLC). My work also covered low-carbon generation, smart-grid management, intelligent mobility and building energy efficiency. The programme concluded in 2026.

VASCULAB: Vascular simulation and haemodynamic analysis

2024

In VASCULAB, conducted with a university hospital, we modelled intravascular flow associated with bypass surgery using computational fluid dynamics. We compared how vessel diameter, flow rate and blood viscosity affect flow and pressure distributions. We interpreted the results in terms of the importance of these parameters when assessing postoperative haemodynamic conditions.

AEROCAS: Wind-turbine power forecasting

2023

In AEROCAS, we developed an ensemble model to forecast short- and medium-term wind-turbine power generation from meteorological and operational data. We first compared KNN, decision trees, Random Forest and XGBoost using wind speed and direction, temperature and turbine operating parameters. We then evaluated an approach combining their outputs. The results indicated its potential to improve the accuracy of generation forecasts.

PREDIAB: Ensemble learning for early diabetes detection

2023

In PREDIAB, we studied ensemble-learning methods for the early detection of diabetes using patient data obtained with a university hospital. We compared Voting, Random Forest, Bagging and Gradient Boosting on the same classification task. Rather than ranking models by overall accuracy alone, we considered precision and recall together, including the significance of missing patients at risk. This allowed us to discuss their potential for clinical decision support in light of trade-offs among performance measures.

HEMERA: Machine-learning classification of cancer cells

2022

In HEMERA, we developed a model with a university hospital to classify cancer cells by type. We compared KNN, support vector machines, decision trees and Naive Bayes to inform the modelling approach. The resulting model achieved high accuracy and precision. We considered what these results suggest for the use of machine learning in support of clinical assessment.

HERA: Energy retrofit of traditional housing

2021

In HERA, we used the Özdoğan House to investigate how to improve the energy performance of traditional housing while preserving its architectural character. We analysed thermal comfort, heat losses and consumption, and considered passive measures such as insulation, material choices and window placement alongside renewable energy and heating/cooling systems. The work identified options for reducing consumption while respecting the constraints of the building's traditional architecture.

ONCAS: Dimensionality reduction for breast-cancer cell classification

2021

In ONCAS, conducted with a university hospital, we examined the effect of data representation on the classification of breast-cancer cells. We established a KNN baseline using features extracted from clinical data. We then transformed those features with principal component analysis (PCA) and neighbourhood components analysis (NCA) and re-evaluated KNN. This let us compare changes in performance attributable to feature representation while retaining the same classification algorithm.

DRIVEFF: Vehicle fuel-consumption forecasting

2020

In DRIVEFF, we developed a regression model to forecast vehicle fuel consumption. We compared Lasso, Ridge, ElasticNet and XGBoost to evaluate linear and nonlinear approaches. Our model produced successful forecasting results. The study showed how data-driven models can be used to examine vehicle energy performance and anticipate consumption patterns.

SART: Open-circuit subsonic wind tunnel

2019

In SART, we designed and built an open-circuit subsonic wind tunnel for aerodynamic experiments. Designed for flow control, pressure measurement and the study of aerodynamic forces, the tunnel was used in undergraduate laboratory sessions. It provided a setup in which students could observe flow behaviour and relate measurements to aerodynamic principles.

RÜZGEM: Vertical-axis wind turbine

2018

In RÜZGEM, we designed and completed the manufacture of a vertical-axis wind turbine intended to contribute on-site power for university laboratory buildings. Operation at low wind speeds and suitability for campus conditions were central design considerations. The completed turbine enabled us to assess renewable-energy design decisions on a physical engineering system.

TARIM: IoT-integrated UAV for smart agriculture

2017

In TARIM, we developed an unmanned aerial vehicle prototype equipped with IoT sensors and autonomous control for agricultural monitoring. We designed the system to collect information on plant health, soil moisture, temperature, irrigation needs and indications of disease. Analysis of these data on a cloud platform was intended to help growers monitor field conditions and make informed decisions.

RESEARCH AREAS

Energy and flow systems: Wind-turbine design and aerodynamics; power forecasting, wake interactions and early fault diagnosis; thermal systems, heat transfer and computational fluid dynamics.

Data-driven engineering: Physics-informed machine learning and artificial intelligence; combined analysis of operational, meteorological and acoustic data; energy and fuel-consumption forecasting.

Sustainable infrastructure: Building energy performance and retrofit; smart cities, IoT and visible light communication; smart agriculture.

Biomedical applications: Computational modelling of haemodynamic flow and machine-learning classification of clinical data.

TECHNICAL SKILLS

Engineering analysis and design: ANSYS, OpenFOAM, MATLAB/Simulink, SolidWorks, AutoCAD, Revit and SAP2000

Programming and data analysis: Python, MATLAB, C#, JavaScript and SQL

Machine learning: scikit-learn, TensorFlow and Keras; basic PyTorch

Software and databases: ASP.NET MVC, Entity Framework, MS SQL Server and Dynamics 365

ACADEMIC DISTINCTIONS

2023 · First in the Mechanical Engineering master's programme, Istanbul Technical University; high honours.

2019 · First in the Civil Engineering Department (double major), Işık University.

2018 · Top graduate of the Faculty of Engineering, Işık University.

2018 · First in the Mechanical Engineering Department, Işık University.

PROFESSIONAL MEMBERSHIPS AND LANGUAGES

Chamber of Mechanical Engineers (TMMOB) and Chamber of Civil Engineers (TMMOB), both with signing authority.

Turkish: native · English: advanced.

Academic references are available upon request.